Module 05 · Lecture 4

Mental models and cognition

How people figure out how things work, and why they get it wrong

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Summary

- **Why look at cognition**? People act on a model, not on the machine
- **Mental models**: how they form, how they run, how they fail
- **Norman's seven stages** of action, and the two gulfs
- **Cognition outside** the head: distributed, external, embodied

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TOPIC 1

Why cognition?

The model in your head is what you act on

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QUESTION FOR THE CLASS

It is cold. You turn the thermostat up to 30 °C so that the room heats faster. Does it?



Two more situations:

- At the crossing, you press the button three times.
- Your laptop is slow, so you reboot it.

Show of hands for each: does it work?

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They all come from a model in your head

What you do	What you believe (your model)	What is actually the case
Turn the thermostat up to 30 °C	*"The higher the setting, the harder it heats."*	Most thermostats are switches. The boiler runs flat out until the set point, then stops. 30 °C only means it stops later.
Press the crossing button three times	*"Each press is a request. More presses, faster green."*	Fixed cycles. Many buttons are not even connected. Example: in 2004 about 2,500 of New York's 3,250 did nothing.
Reboot the slow laptop	*"A restart cleans whatever went wrong."*	Often true, for the wrong reason. The successes reinforce a model that also sends people to reboot the projector.

We act on a model of how the thing works, not on the thing itself. That model is the subject of this module: where it comes from, how it runs, and how the interface can shape it.

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Why look at cognition?

Our goal is to understand how people **reason** about an interface and **react** to it.

A **cognitive framework** helps because:

- Theoretical models have predictive and explanatory power: they say what a user will try, and why it failed
- We can borrow theories of cognition developed elsewhere (psychology, ergonomics) instead of guessing

Internal frameworks: inside the head

Mental models: the user's idea of how the thing works

The gulfs of execution and evaluation (Norman)

Information processing: the human as a processor with stages

External frameworks: outside the head

Distributed cognition: across people, artefacts and time

External cognition: using the world (notes, maps, post-its) to think

Embodied interaction: the body and the social setting shape meaning

In the project: the COL operator already has a model of the river. Your panels either fit it or must visibly correct it. Ignoring it is not an option.

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TOPIC 2

Mental models

What they are, how they form, how they fail

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What is a mental model?

"In interacting with the environment, with others, and with the artifacts of technology, people form internal, mental models of themselves and of the things with which they are interacting."

Donald Norman, in Gentner and Stevens (eds.), Mental Models, 1983

People use their mental model to:

- **Reason about a system**
  - how it works, and how to interact with it
- **React to a situation**
  - figure out what to do when things go wrong

Same symptom, two models, two actions

The Wi-Fi is slow.

- Model A: "the router gets tired." Action: reboot the router.
- Model B: "the neighbours are on the same channel." Action: change the channel.

You never see the model. You only see the actions it produces. The interface is the only place where you can correct it.

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Three models, and only one channel between them

Designer's model
How the designer thinks the product works and should be used.

→

System image
Everything the product shows: screens, labels, sounds, behaviour, the manual nobody reads.

→

User's model
What the user concludes from using it. Built from the system image and from past experience.

no direct channel: the designer never meets the user

- The designer **expects the user to build the same model**. But the only message the designer sends is the product itself.
- When the system image hides or contradicts the mechanism, the user's model is wrong, and the designer calls it "user error" (Module 04).
- **Conceptual design is the design of the system image, so that the user's model is close enough to predict what the system will do.**

In the project: you will never sit next to the operator at 3 a.m. on a flood night. Only your two screens and the wall will be there to explain themselves.

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What informs the user's model: seven design concepts

Visibility	**Can I see the state, and the controls?**	A fuel gauge is visible. The battery level of a wireless mouse is not, until it dies.
Affordance	**What the object offers to be done with it**	A handle affords pulling, a flat plate affords pushing. Norman doors get it backwards (Module 04).
Mapping	**The relation between controls and effects**	Four burners in a square, four knobs in a row: which knob is which?
Constraints	**What cannot be done**	USB-C fits either way. USB-A needed three attempts. A gate cannot open beyond 100 %.
Transfer effects	**What past use of similar objects brings**	Every phone puts "back" on the same edge. Change it and users swipe into the void.
Cultural associations	**What the culture already means by it**	Red means stop or danger. On Chinese stock screens, red means the price went up.
Feedback	**What tells me what just happened**	The click of the key, the beep of the ATM, the spinner. Late feedback means a second press.

All seven inform the user's mental model. None of them is the mechanism itself: they are the system image.

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Example: forming a model just by looking



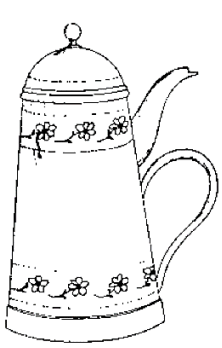
- **Affordance:** useful for cutting… and it can hurt if not handled with care
- **Mapping:** holes for fingers are suggested, and constrained, by their shape
- **Constraints:** big hole for several fingers, small hole for the thumb
- **Positive transfer and cultural idiom:** used since childhood, the mechanism never changed
- **Mental model:** the physical object implies how the operating parts work

Question for the class. The model is not perfect. Can a left-handed person use them?

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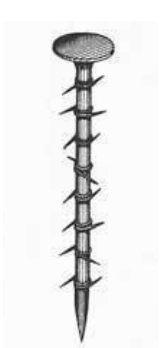
Objects that lie

ACTIVITY · 2 min











Task: Each object promises something. Say what each object promises and where the promise breaks.

Jacques Carelman, Catalogue d'objets introuvables, 1969

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Draw a bicycle

ACTIVITY · 5 min

Four volunteers, one (black)board

- Two who have not ridden a bicycle in the last five years, two who ride regularly
- Draw, from memory, a bicycle that would work: frame, pedals, chain, wheels, steering
- Three minutes. No phones
- The class judges: would it move? Would it steer?

What usually happens

About 40 % of adults make a basic error: the chain on the front wheel (it could not steer), a frame tube joining both wheels.

Regular cyclists do only a little better (Lawson, 2006).

Why it matters

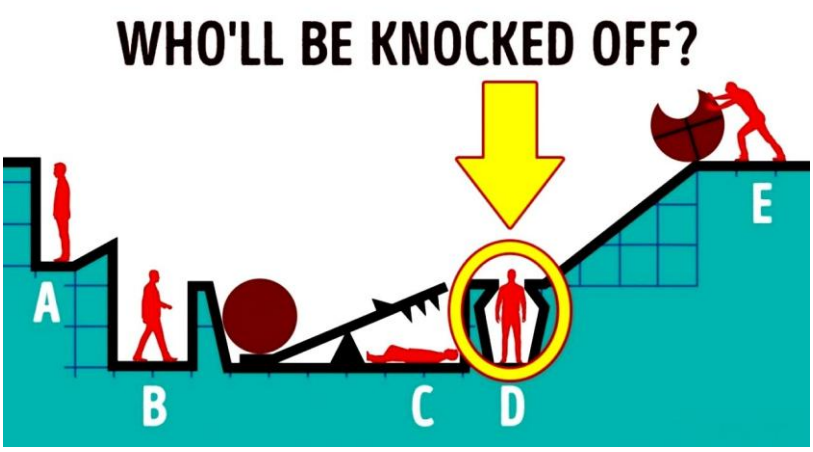
Before drawing, everyone was confident. This is the illusion of explanatory depth (Rozenblit and Keil, 2002).

Mental models are incomplete, and we do not know it until we run them.

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Run your model: who gets knocked off?

ACTIVITY · 1 min



Vote: A, B, C, D or E

Thirty seconds. Then hands up.

What you just did

You ran a model of physics in your head and predicted an outcome, without touching anything.

That is what users do with every interface, before every click.

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Models are runnable, and they are imperfect

Runnable

The user perturbs the system to figure out how it works and builds a notion of causality: “doing this will result in that.”

Then the model runs in two directions:

Prediction: what will happen if I do this? Which action should I choose? (the intention side)

Explanation: why did the system respond like that? (the evaluation side of Norman's seven stages)

Every click is:

- *preceded by a prediction*
- *followed by an explanation.*

Characteristics of mental models

- **Incomplete**: the bicycle with the chain on the front wheel
- **Unstable**: details are forgotten, the model evolves with use
- **Not accurate**: the fridge (coming up)
- **Simplified**: a complex phenomenon reduced to what fits in a head
- **Superstitious**: the thermostat at 30 °C, the crossing button
- **Expressible as if-then rules**: “if I press this, then that lights up”

Norman, 1983

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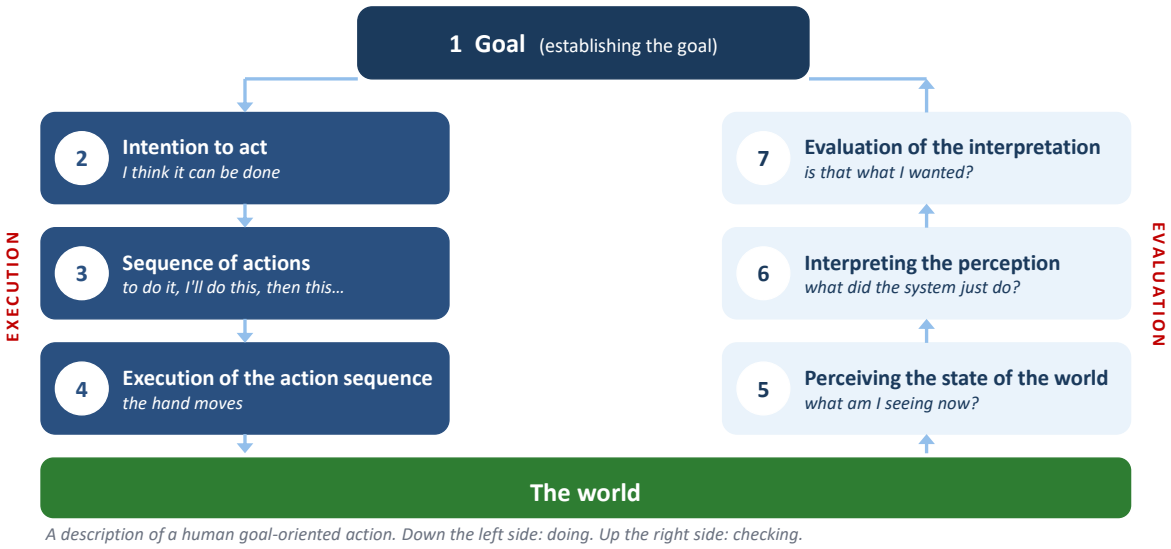
TOPIC 3

Norman's seven stages

A description of goal-oriented action, and the two gulfs that separate you from the machine

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Norman's seven stages of action



A description of a human goal-oriented action. Down the left side: doing. Up the right side: checking.

Norman, The Design of Everyday Things (1988, 2013)

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The seven questions the user asks

Intention side (execution)

1. **What do I want to do?** *the goal*
2. **I think it can be done. How?** *intention*
3. **To do it, I'll do this, then this…** *the sequence*
4. **(I physically check the mental model)** *execution*

Interpretation side (evaluation)

5. **What am I now seeing?** *perception*
6. **What did the system just do?** *interpretation*
7. **Is that what I wanted? What did I do to make it do that?** *evaluation*

These are the questions of the cognitive walkthrough. Module 06, next in this session. In Phase 1 of the project you will ask them, on paper, about another group's sketches.

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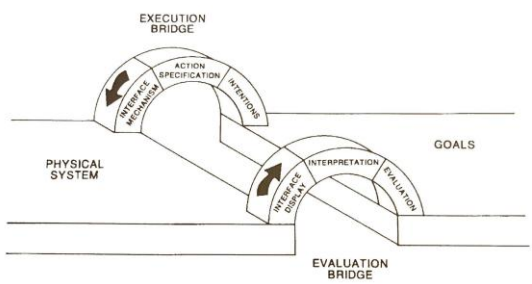
The seven stages at the COL: one mission

	what the operator does	what the panel must offer
1 Goal	Raise the release at Touvedo from 80 to 120 m³/s	The mission is visible on the wall, with the current value next to it
2 Intention	"I use the gate controls of Touvedo"	Visibility: the controls exist and the operator can see where
3 Sequence	Select Touvedo, select the gate, set 120, confirm	Each step visible. Constraints: *there is no way to type 1200 in the interface*
4 Execution	Drag the slider, or type, or turn a knob	Affordance: *the control looks like what it does*
5 Perceive	Something changes on the screen	Feedback within a tenth of a second (Module 09), even if the gate takes minutes
6 Interpret	"The gate is opening, the flow will rise"	Show command and state separately: *ordered 120, currently 92*
7 Evaluate	"Did I reach 120? Is the level downstream rising as expected?"	The graph, with the target line, and the expected time to reach it

Third column: your first checklist. It is also how the cognitive walkthrough of Phase 1 will read your sketches.

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The two gulfs



Source: User centered system design: new perspectives on human-computer interaction by D Norman. Copyright 1986 by Taylor & Francis Group LLC - Books. Reproduced with permission of Taylor & Francis Group LLC.

Design bridges both gulfs: controls that match intentions and displays that answer questions.

Norman, in User Centered System Design (1986). Taylor & Francis

Gulf of execution: "How do I do it?"

The distance between what I want and the actions the system offers.

The projector: remote with two buttons (Module 04).
A gate control buried in a submenu.

Gulf of evaluation: "What happened?"

The effort to work out the state of the system from what it shows.

The crossing button with no light.
"Saving…" with no progress.
A gate that takes three minutes and shows nothing.

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When do the seven stages apply?

Best for exploratory learning

First use, or a new situation. The user has no strong expectation, so every stage is conscious and can fail.

The first day at the COL. The first flood.

Less useful for automatic behaviour

Highly learned, semi-automatic tasks: the user already expects what will happen, and the stages collapse into tiny, fast loops.

Driving to work. A normal Tuesday at the COL.

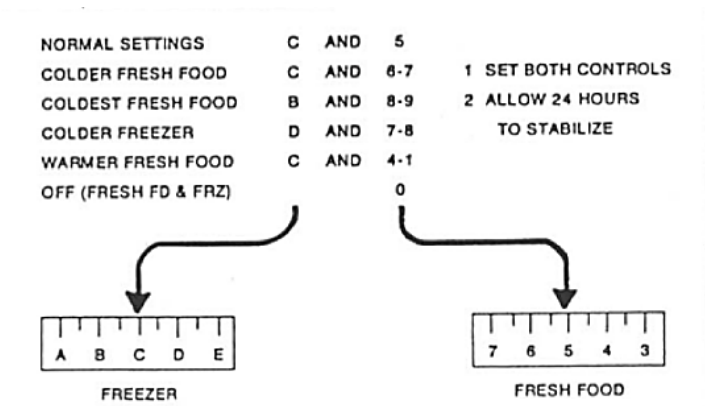
…unless something unexpected happens. Then all seven stages come back, in full: the low-fuel warning on the motorway, the first alarm of a flood night. The same screen has to serve both days (the "variability" of Module 04).

Method consequence: the cognitive walkthrough (Module 06) evaluates first use, stage by stage.
Usability tests (Module 11) look at repeated use and time.

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The fridge

QUESTION · 1 min



What the panel says

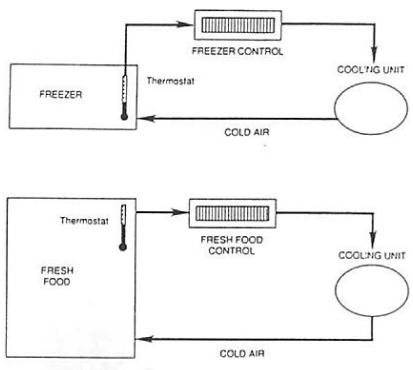
Two controls, one labelled FREEZER, one labelled FRESH FOOD. A table of settings and an instruction: set both, allow 24 hours to stabilise.

Question for the class. The freezer is too warm. The fresh-food compartment is fine. What do you set?

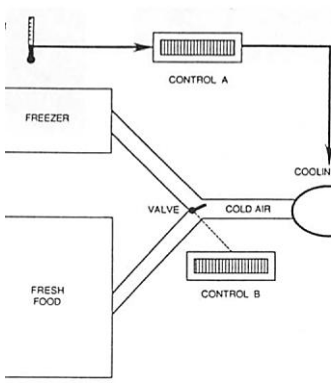
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The fridge: the user's model and the real model

The user's model: two units, independent



The real model: one unit, two interacting controls



Why it fails

A sets the compressor and B splits the cold air. Change one, both compartments change.

Nothing on the panel says so.

Both gulfs at once. Execution: which knob, by how much? Evaluation: wait 24 hours to find out. **This is not user error**. The designer hid the mechanism.

Norman, The Design of Everyday Things

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The third internal framework: information processing

Input	Encoding	Mental comparison	Response selection	Response execution	Output
stimuli: light, sound, touch	combine the stimuli and make them intelligible	search memory for something similar	choose a reaction, from what you know	do it	the response: voice, motion

- Metaphor: a human is like an information processor
- There are ordered processing stages, each with input and output
- Within each stage, processes (comparing, matching) act on mental representations
- **Useful for making predictions and identifying bottlenecks: how to check it was done; what is the capacity?**

Leads to the Model Human Processor

Perception, cognition and motor action as three processors with memories and cycle times. (With numbers in Module 09)

For coffee, not for the exam

Computation is state change. Is cognition a series of state changes?

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Cognitive load and attentional load

Question for the class. Which costs more: following this lecture for two hours, or understanding one dense slide? Are they the same kind of cost?

Load

The amount of mental resources it takes to process information.

Cognitive load

How much "brainpower" a task takes: understanding, deciding, computing.

The dense slide. A release decision on a flood night.

Attentional load

How much attention a task takes: monitoring, noticing, staying on it.

Two hours of lecture.

Twelve screens on a wall, all day.

Design rule (tip): keep the attentional load of the wall low (few things move, few things blink) so that the operator's cognitive capacity is intact when the decision comes. Module 09 puts numbers on both.

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TOPIC 4

Cognition outside the head

Distributed cognition, external cognition, embodied interaction

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Three external frameworks

Distributed cognition

Cognition happens across individuals, artefacts and time. The unit is **the crew**, **the control room**, **the team**, not one head.

The cockpit: the altitude lives in the radio call, the dial, the pilot's hands and the captain's glance.

External cognition

Using external representations (books, maps, diagrams, lists, post-its) to work with your internal ones.

A shopping list. A whiteboard. The long multiplication you cannot do in your head.

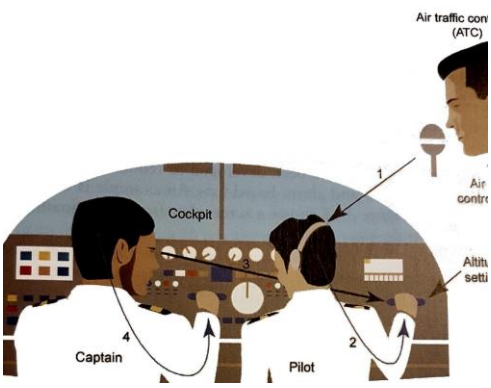
Embodied interaction

Meaning is shaped through physical and social interaction with the world. Embodied = "incorporado".

Pointing at the wall so that everyone sees what you see. Stop at a given time to think and make other see you.

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Distributed cognition: how a cockpit remembers



Propagation of representational states

1. ATC gives clearance to fly to a higher altitude *verbal*
2. The pilot changes the altitude meter *mental and physical*
3. The captain observes the pilot *visual*
4. The captain flies to the higher altitude *mental and physical*

- How people work together to solve a problem, and how knowledge is shared and accessed
- Verbal and non-verbal behaviour: what is not said is implied by glances and gestures
- The coordinating mechanisms: rules, procedures, checklists

At the COL: where is the last release order stored? In the operator's head, on the screen, in the logbook, in the phone call to Touvedo? Your interface decides.

Hutchins, How a cockpit remembers its speeds, 1995

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External cognition: three mechanisms

TRY IT · 30 s

Externalising to reduce memory load

Diaries, calendars, to-do lists, alarms.

They remind you to do something, what to do, and when.

Computational offloading

A tool plus a representation carries part of the computation.

3 × 2 in your head. 234 × 456 in your head. 234 × 456 on paper. With a calculator. And CCXXXIV × CDLVI.

Same problem, four difficulties. The representation changes the task.

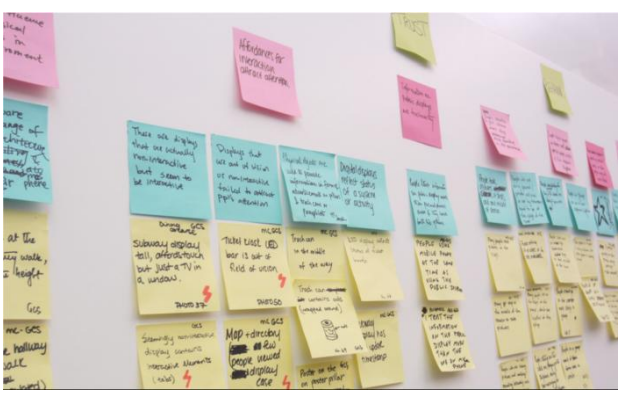
Annotation and cognitive tracing

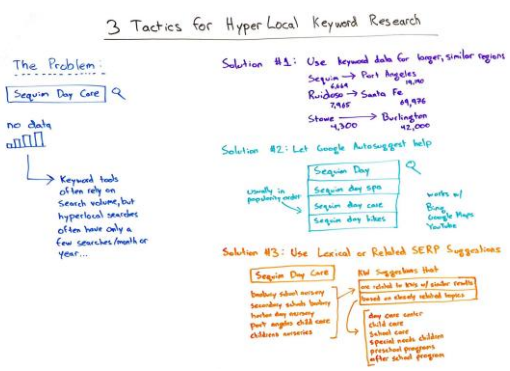
Annotation: crossing items off, highlighting, circling.

Tracing: moving items into new orders or piles, like a wall of post-its.

Try it. 234 × 456 in your head. Now CCXXXIV × CDLVI, on paper. Which representation made the task hard?

Your project runs on external representations





Session S4: post-its of three colours, expand then contract. **Session S5:** storyboards. **All semester:** the sketchbook.

These are not paperwork. They are where the group thinks, where its memory lives, and what the walkthrough of Phase 1 will read. A group whose ideas are only in heads has no shared model. The wall is the model.

Embodied interaction

Understanding interaction in terms of **practical engagement** with a social and a physical environment.

It came from **watching how people get things done** in real situations, not in the lab.

Dourish, Where the Action Is, 2001

At the COL: the wall is public, the two screens are private. Decide what goes where with that in mind: what must the whole room see at once, and what is one operator's business?

The emotional quality of interactions
A device that feels hostile is used less, and worse. The gym watch of Module 04.

Publicly available actions in shared spaces
When the operator stands up and points at the wall, the whole room knows something is happening.

The body mediates the interaction
A big red button and a menu item do the same thing. They are not the same action. Kinect, Module 04.

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Wrap-up

Mental models
what they are, how they form from the system image, how they run (explain, predict) and how they fail (incomplete, inaccurate, superstitious)

Norman's seven stages and the two gulfs
goal, intention, sequence, execution, perception, interpretation, evaluation. Execution: how do I do it? Evaluation: what happened?

Information processing and load
stages with inputs and outputs. Cognitive load vs attentional load. Numbers in Module 09

Cognition outside the head
distributed (the cockpit, the control room), external (lists, post-its, the whiteboard), embodied (the body and the shared space)

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NEXT

Module 06: the cognitive walkthrough

The seven questions, asked on paper, about another group's sketches. Phase 1 of the project, in November.